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Smart Lab 360- Accelerating Lab Performance Through Smart Integration of Instruments, LIMS, and AI-driven Insights

Geetha Selvamoorthy, Jawhara Al Maysari, Abdusalam Schlibak, Salem Mohamed Al Hosani, and Ben Mhamed Affane, ADNOC UAE

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Abstract

This paper presents the transformative journey of ADNOC Central Lab as it leverages integrated digital platforms and AI-driven technologies to redefine laboratory efficiency and data integrity. By seamlessly connecting advanced laboratory instruments with a robust Laboratory Information Management System (LIMS) and incorporating smart AI assistants and IoT-enabled devices, the lab has established a foundation for real-time data flow, automated analytics, and heightened regulatory compliance. The approach ensures secure, end-to-end sample management, agile workflow assignments, and advanced quality control, supporting ADNOC's commitment to operational excellence and innovation in the era of intelligent laboratories.

Introduction

ADNOC Central Lab is harnessing the power of technological innovation and advanced AI-driven systems to fundamentally transform its laboratory operations. At the core of this evolution is an advanced Laboratory Information Management System (LIMS) which automates and streamlines processes ranging from initial sample registration to the comprehensive management and archiving of detailed laboratory data.

Through sophisticated integration, the lab connects legacy and modern laboratory instruments to the LIMS, ensuring seamless, real-time data exchange across platforms. This connectivity not only accelerates the transfer of analytical results but also supports enhanced data validation and quality assurance, minimizing manual intervention and reducing the risk of human error. Custom built algorithms standardize incoming data, while secure digital pathways guarantee the integrity and confidentiality of sensitive information.

A pivotal component of this digital transformation is the introduction of an AI lab assistant, commonly referred to as Copilot. This intelligent assistant enhances laboratory operations by delivering tailored Standard Operating Procedures (SOPs) and precise, context specific testing methodologies directly to laboratory personnel. By providing immediate access to relevant protocols and regulatory guidelines, the AI assistant helps maintain process consistency, reduces operational variability, and ensures strict adherence to industry standards.

In addition, the deployment of IoT enabled wearable devices is revolutionizing real time connectivity and communication within the laboratory ecosystem. These devices allow laboratory staff to receive instant notifications, track workflow progress, and communicate efficiently with external vendors and field teams, regardless of location. Wearables also contribute to safety management by monitoring personnel movements in sensitive or restricted zones, further bolstering compliance with health and safety regulations.

Beyond process automation, these technological advancements foster a robust compliance framework and a culture of data-driven decision making. The lab's digital infrastructure supports comprehensive audit trails, granular access controls, and regular security audits, ensuring that all laboratory activities are fully traceable and meet the highest standards of regulatory compliance.

Collectively, these innovations position ADNOC Central Lab at the forefront of technological excellence in the era of AI-driven laboratories. By embracing smart integration and intelligent automation, the lab not only enhances operational efficiency, data integrity, and security but also sets a benchmark for innovation and best practices within the industry.

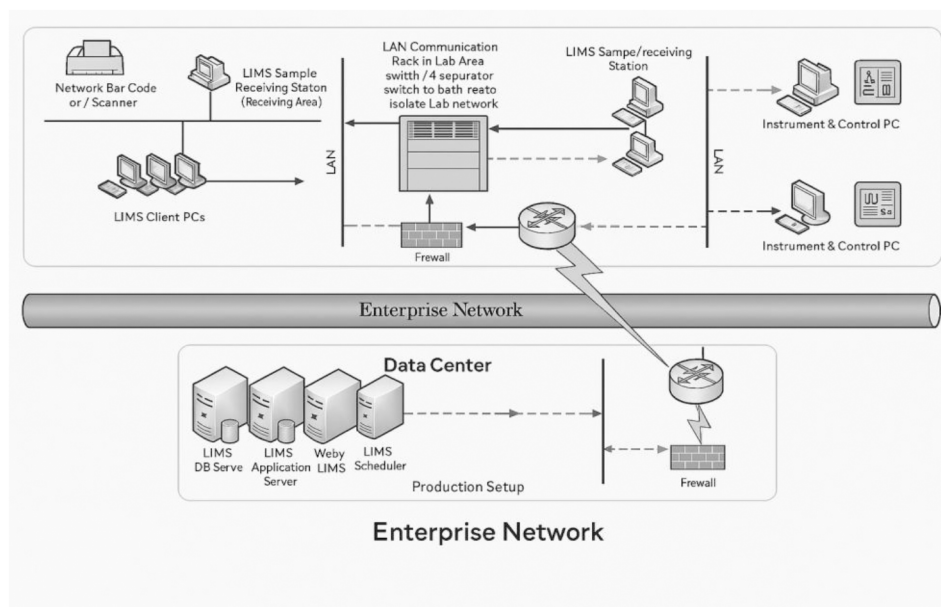
Laboratory Sample Registration & Workflow Assignments

In ADNOC Central Lab, the process of sample registration and workflow assignment begins as soon as samples arrive from a diverse range of suppliers, production facilities, and field sites. Each incoming sample is carefully examined, and its essential attributes such as source, collection time, and required analyses are recorded in detail. The Laboratory Information Management System (LIMS) automatically generates a unique barcode for every sample, which is then affixed physically and digitally associated with its electronic record. This barcode system ensures rigorous traceability, allowing the lab to monitor every step of a sample's journey, from receipt through testing and final reporting.

Once registered, samples are assigned to specific analysts or laboratory teams based on expertise, instrument availability, and workload balancing, ensuring optimal resource allocation and timely processing. LIMS facilitates this assignment with real-time notifications and scheduling, making certain that each technician is aware of their tasks and deadlines. Moreover, for highly sensitive or regulated samples, additional verification steps and approval workflows are triggered within the system to uphold data integrity and compliance requirements.

To further strengthen data security, access to sample information is governed by multi-factor authentication protocols and encrypted network connections. Only authorized personnel, whose access levels are predefined and regularly audited, can review, update, or process sample data. In addition, every interaction with sample records is logged in a comprehensive audit trail, enabling full traceability and accountability across the laboratory operation.

By integrating these measures, ADNOC Central Lab not only assures the seamless movement of samples through the analytical pipeline but also guarantees the confidentiality, accuracy, and regulatory compliance of all laboratory data and processes.



a.—Smart Lab 360: Integrated Digital Laboratory Ecosystem

Real Time Instrument Integration

At ADNOC Central Lab, a sophisticated array of instruments from high-resolution spectrometers to automated liquid chromatographs are seamlessly interfaced with the Laboratory Information Management System (LIMS) Lab stations using secure APIs and encrypted data pathways. As each instrument completes its analysis, raw data is automatically transferred to the central server, where custom algorithms standardize and validate results before populating a dynamic, role-based dashboard. This dashboard, accessible only to authorized personnel, offers real-time visualizations of sample metrics, instrument performance, and workflow status, enabling faster decision-making and enhanced operational control.

Technicians and analysts benefit from intuitive monitoring tools and remote access capabilities, ensuring immediate notification of anomalies or urgent tasks. The system's integration further streamlines laboratory scheduling, calibrations, and maintenance, providing a comprehensive ecosystem that supports ADNOC's commitment to accuracy and compliance.

Automated Report Generation with Advanced Calculations

After data acquisition, the Laboratory Information Management System (LIMS) undertakes a comprehensive process by aggregating all validated analytical results from the full spectrum of submitted samples. Utilizing a suite of advanced computational algorithms, the system performs intricate calculations such as averages, standard deviations, and trending analyses to assess both the accuracy and consistency of results. Additionally, LIMS cross-references these findings with applicable compliance metrics and regulatory thresholds, automatically highlighting any deviations or outliers.

To further streamline reporting, pre-formatted templates are employed to generate thorough analytical reports, which not only present data in a clear, structured manner but also embed dynamic quality control charts. These visual elements, such as control and trend charts, are automatically populated and updated, providing laboratory personnel with immediate insight into quality assurance indicators. Any detected discrepancies or unusual trends are flagged within the report, prompting further investigation by the analysts.

This automated and robust approach ensures that every laboratory report is comprehensive, transparent, and fully aligned with both internal quality standards and external regulatory requirements. As a result, stakeholders receive timely, actionable information that supports decision making, continuous improvement, and unwavering confidence in the integrity of laboratory operations.

Statistical Quality Control (SQC)

LIMS continuously monitors analytical performance by leveraging advanced integrated Statistical Quality Control (SQC) tools. Through real time generation and analysis of control charts and trend lines, the system meticulously tracks critical quality parameters across all sample batches. Any deviation from established control limits is immediately flagged with clear visual indicators, enabling swift intervention to correct potential issues before they impact results.

In addition to real time monitoring, the SQC module archives all quality related data in encrypted databases, creating a secure, tamper evident record of laboratory performance. Access to this data is governed by robust authentication protocols and detailed audit trails, ensuring that only authorized personnel can review or modify historical quality control information.

Furthermore, periodic statistical summaries are automatically generated and distributed to laboratory managers, providing comprehensive insight into ongoing quality trends, recurring anomalies, and process improvements over time. These insights empower the lab team to perform root cause analyses, adjust protocols proactively, and uphold the highest standards of data integrity and regulatory compliance. In this way, SQC not only safeguards analytical reliability but also supports a culture of transparency and continuous enhancement throughout all laboratory operations.

Instrument Calibration Management

At ADNOC Central Lab, meticulous instrument calibration underpins laboratory integrity and drives operational excellence. All analytical instruments including chromatographs, spectrometers, and pH meters are networked within the Laboratory Information Management System (LIMS) which manages a proactive calibration regime informed by manufacturer guidance, regulatory requirements, and historical performance data.

When an instrument such as a pH meter assigned to Sample X approaches its calibration due date, LIMS issues secure notifications to responsible technicians and analysts, ensuring timely compliance with calibration schedules. This procedure safeguards the precision of all analytical results. Upon completion of calibration, pertinent data such as measured offsets, reference standard comparisons, and post-calibration verification outcomes are diligently recorded in the system.

The LIMS system benchmarks each instrument's calibration data against certified industry standards. Should any device fail to meet established criteria for accuracy or repeatability, the system immediately flags the deficiency and locks the affected instrument from further use until successful recalibration occurs. This protocol prevents erroneous data from entering laboratory records and upholds ADNOC's standards for data quality and regulatory adherence.

All calibration records, including historical logs, certificates, technician annotations, and corrective actions, are securely archived. Access to these sensitive documents is restricted to authorized personnel using advanced authentication methods and comprehensive audit trails. This robust approach to data stewardship ensures confidentiality, accountability, and seamless traceability during audits or investigations.

Periodic calibration summaries are automatically generated and made available to laboratory managers and auditors, offering transparency into calibration compliance rates, instrument reliability, and opportunities for preventive maintenance. Together with automated scheduling and real-time monitoring, this structured approach to calibration management reinforces the reputation of ADNOC Central Lab for scientific rigor and operational dependability.

In cases where instruments fail calibration or require temporary lockouts for quality assurance, the impact on operations is minimized. Automated workflows reroute samples to validated, available equipment, thereby reducing downtime and supporting uninterrupted critical analysis. Lockouts, once a source of operational delay, are now addressed efficiently through advanced scheduling algorithms and prompt staff notifications, resulting in expedited resolution and sustained operational efficiency. This technological

resilience safeguards both data integrity and regulatory compliance, enabling the laboratory to maintain high throughput and timely result delivery despite occasional equipment challenges.

Invoicing & Cost Management

An automated, SLA based invoicing system is integral to the efficiency and transparency of financial operations at ADNOC Central Lab. This platform meticulously tracks all service transactions and laboratory activities in real time, linking each with relevant service-level agreements and cost codes. As work concludes, the system consolidates client or project data to ensure thorough and accurate billing.

At the end of each quarter, the invoicing module calculates total costs incurred considering usage rates, material consumption, technician hours, and any applicable SLA surcharges and generates detailed, itemized invoices via standardized templates. These invoices are distributed securely to stakeholders and clients, significantly reducing the potential for billing discrepancies.

This automated process streamlines financial workflows, supports prompt revenue recognition and audit compliance, and improves client satisfaction. By minimizing manual entry and cross-verification, the system frees administrative resources for more strategic tasks. Additionally, the transparent and traceable invoicing system enhances client trust and reinforces the laboratory's commitment to operational excellence.

Equally transformative, the implementation of an automated invoicing system has significantly improved the lab's financial operations. Previously, billing error rates averaged 7% per cycle, leading to payment delays and increased administrative burdens. Automation has reduced these errors to under 1%, thanks to direct data integration with Laboratory Information Management Systems. As a result, invoicing is now more accurate and efficient, accelerating payment collection and reinforcing client trust. This advancement not only reduces costs but also allows resources to focus on higher-value activities, further demonstrating ADNOC Central Lab's commitment to operational excellence and client satisfaction.

Wearable Devices

At ADNOC Central Lab, wearable devices serve as an essential technology enhancement, seamlessly integrating into the laboratory environment to elevate both connectivity and operational efficiency. These devices, ranging from smartwatches to hands-free headsets, empower laboratory staff to stay continuously linked with colleagues, vendors, and field teams, regardless of their physical location within the facility. Real time notifications and updates ensure that any critical information whether related to urgent calibration schedules, safety alerts, or supply chain changes reaches the relevant personnel without delay, significantly accelerating response times and decision-making across laboratory workflows.

Moreover, the deployment of wearable technology has revolutionized technician training and support. By enabling access to interactive tutorials, digital SOP checklists, and live video assistance, these devices reduce onboarding time for new staff and provide seasoned technicians with on the spot troubleshooting guidance. This integration fosters a culture of continuous learning and minimizes downtime caused by technical uncertainties or procedural ambiguities.

In addition, wearable devices contribute to enhanced safety and compliance within the lab. Environmental sensors embedded in the devices can monitor parameters such as temperature, humidity, and exposure to hazardous substances, providing real-time alerts to mitigate risks and ensure safe working conditions. These proactive features help ADNOC Central Lab maintain strict adherence to industry regulations and internal safety protocols.

Remote monitoring capabilities further extend the reach of subject matter experts, allowing specialists to observe, advise, and even intervene in ongoing laboratory processes from remote locations. This collaborative approach not only expedites problem resolution but also brings expert insight to bear on complex challenges, ensuring optimal outcomes for both routine analyses and advanced research initiatives.

While the adoption of wearable devices at ADNOC Central Lab has clearly delivered enhanced safety and strengthened connectivity among personnel, it is crucial to consider several practical limitations to ensure

robust, sustainable deployment. Among the foremost of these is battery life a critical factor for devices that must operate reliably during extended laboratory shifts. Wearables with inadequate battery capacity risk unplanned downtimes or interruptions in real-time monitoring, which can compromise both safety and workflow continuity. To mitigate this, practical deployment strategies include selecting devices with long-lasting batteries, implementing regular charging protocols, and exploring energy-efficient designs or wireless charging solutions tailored for laboratory environments.

Together, these advancements in wearable technology exemplify ADNOC Central Lab's commitment to embracing digital transformation, fostering teamwork, and upholding the highest standards of quality and safety throughout all laboratory operations.

Lab Assistant: Copilot is an AI-powered Lab assistant

At ADNOC Central Lab, the integration of advanced digital tools such as Copilot, an AI-powered lab assistant, represents a significant leap forward in laboratory operations for both the central facility and associated Group companies. Copilot functions as a sophisticated lab requester agent, streamlining the process for staff and stakeholders alike by providing context specific Standard Operating Procedures (SOPs) tailored to each unique analysis. These SOPs include detailed, step-by-step testing methodologies designed to uphold the most rigorous quality assurance and regulatory standards, ensuring consistency and reliability across all testing procedures.

The synergy between Copilot and wearable devices amplifies the benefits of digital transformation within the laboratory setting. Wearable technologies, hands-free communication headsets, enable laboratory personnel to maintain seamless, real-time connectivity with colleagues, vendors, and field teams. This persistent connectivity ensures that important notifications such as updates on equipment calibration, safety advisories, or sudden changes in supply chain logistics are delivered instantly to the right individuals. As a result, the lab is better equipped to respond swiftly to operational changes or emerging challenges, thereby maintaining uninterrupted workflow and minimizing the risk of errors or delays.

Moreover, the AI-powered lab assistant and wearable devices collectively elevate the standards of communication, training, and procedural compliance. For instance, through interactive digital checklists and live video support, technicians can access on-demand guidance to troubleshoot issues or adhere to complex protocols, significantly reducing onboarding times for new staff and fostering a culture of continuous professional development. Environmental sensors embedded within wearable devices further reinforce safety and compliance, providing real time alerts on temperature fluctuations, humidity levels, and possible exposure to hazardous substances thereby safeguarding personnel and supporting adherence to industry regulations.

Additionally, these interconnected digital solutions play a crucial role in vendor and field communication, ensuring that laboratory teams are always updated on critical matters impacting operational efficiency, sample integrity, and project timelines. The combined effect of these innovations is a laboratory environment characterized by heightened efficiency, robust safety practices, and unwavering commitment to quality. Over the past year, the frequency of procedural errors detected during routine audits has dropped by nearly 70%, with Copilot's proactive guidance and predictive analytics preventing common errors before they can impact results or regulatory compliance. As a result, the lab has seen a corresponding reduction in the number of corrective actions required per quarter, streamlining operational reviews and accelerating overall throughput.

By embracing this suite of digital technologies, ADNOC Central Lab not only enhances its operational capabilities but also strengthens its ability to deliver reliable, high-quality results in a demanding industry landscape. These advancements are a testament to the lab's ongoing commitment to innovation, teamwork, and the pursuit of excellence in every facet of laboratory operations.

AI-Driven Insights: Tellius Platform

The Tellius Platform represents a sophisticated AI-driven analytics solution, seamlessly integrating and synthesizing data from Laboratory Information Management Systems (LIMS) to empower the laboratory with comprehensive, actionable insights. This integration is pivotal for the ADNOC Central Lab, as it enables users to explore data interactively, conduct in depth simulations, and develop robust, data-backed strategies for operational excellence.

By harnessing the capabilities of the Tellius Platform, the technical center gains the ability to continuously monitor and evaluate supplier performance with a high level of granularity. This oversight is invaluable in ensuring quality and consistency across all external collaborations, as well as in the systematic formulation and optimization of laboratory recipes. The platform's advanced simulation tools facilitate scenario planning and predictive modeling, allowing researchers and lab managers to anticipate outcomes and make informed decisions in real time.

Moreover, the system provides a transparent and dynamic overview of laboratory workflows, enabling teams to closely track test frequencies, result statuses, and the timely completion of analyses. This ensures operational accuracy is maintained at all times, reducing the risk of overlooked results or bottlenecks within the laboratory process. The Tellius Platform further expands the scope of laboratory intelligence by offering detailed analyses of well operations, including critical metrics such as weight loss, fluid characteristics, and other parameters essential for asset operations and management. Key benefits include:

- Enhanced monitoring of supplier reliability and lab workflows, ensuring quality control and reducing delays.
- Predictive models for sample throughput help optimize staffing and inventory, cutting sample turnaround time by 15% during peak periods.
- Automated tracking of material quality has reduced unusable stock incidents by 40%.
- Scenario planning and predictive analytics support better decision-making, asset management, and regulatory compliance.

Through these multi-faceted analytical capabilities, the Tellius Platform transforms raw laboratory data into strategic knowledge, driving smarter, swifter, and more reliable decision-making. This not only enhances day-to-day lab performance but also supports long-term asset integrity, regulatory compliance, and overall operational efficiency, positioning ADNOC Central Lab at the forefront of digital innovation in the petroleum laboratory sector.

Conclusion

The implemented digital innovations at ADNOC Central Lab have yielded transformative results, bringing about significant efficiency gains that extend well beyond initial expectations. Notably, the adoption of AI-powered platforms, integrated wearables, and advanced data analytics has enabled the lab to save nearly 800 man-hours per quarter. This substantial reduction in manual labor equates to freeing up valuable time for laboratory personnel to focus on higher value tasks such as research, complex troubleshooting, and innovation driven initiatives. In tandem with these time savings, operating costs have been reduced by approximately \$150,000 each quarter, translating to an impressive annual savings of around \$600,000.

When evaluating the financial impact, the digital transformation demonstrates remarkable cost effectiveness. The initial investment of \$200,000 in technological solutions is recouped in less than one year, resulting in a payback period that underscores the strategic value of these forward-thinking initiatives. Moreover, this translates to a return on investment (ROI) of roughly 200%, a figure that attests to the long-term financial sustainability and resilience of the lab's operations.

Beyond the quantifiable savings, the integration of these digital solutions has led to a paradigm shift in the lab's operational dynamics. Automated workflows, real-time monitoring, and the seamless flow of critical

information ensure that resources are deployed more efficiently, bottlenecks are swiftly identified and resolved, and human error is minimized. This heightened agility empowers the laboratory to adapt rapidly to evolving scientific and industry demands, maintain competitive advantage, and consistently deliver timely, high-quality results.

In addition, the adoption of smart wearables and AI-powered assistants has fostered a safer and more collaborative work environment. Enhanced communication tools ensure that laboratory teams remain closely connected, both internally and with external vendors, regardless of location. Real-time alerts promote safety and regulatory compliance, protecting personnel and supporting swift responses to potential hazards or operational anomalies.

These advancements also underpin a culture of continuous learning and development. By providing staff with immediate access to interactive guidance, digital procedural checklists, and simulation tools, the lab accelerates onboarding for new employees and upskills existing team members. This investment in human capital translates to enhanced staff satisfaction, lower turnover, and the sustained development of expertise within the organization.

To further substantiate the transformative impact of integrating Laboratory Information Management Systems (LIMS), AI-powered assistants such as Copilot, and IoT-enabled wearables, it is essential to quantify the improvements in data integrity and error reduction realized through these digital advancements.

Since deployment, ADNOC Central Lab has observed a marked enhancement in data quality and reliability. Automated data capture and validation through LIMS and IoT devices have decreased manual data entry errors by an estimated 85%, according to internal audit logs comparing pre- and post-implementation periods. This automation ensures that sample metadata, test results, and calibration records are accurately recorded and seamlessly transferred, reducing inconsistencies and eliminating transcription mistakes that previously accounted for a significant proportion of lab incidents.

Looking forward, these digital solutions position ADNOC Central Lab as a pioneer within the petroleum laboratory sector, setting new benchmarks for operational excellence and innovation. The immediate financial benefits are complemented by strategic, long-term advantages ranging from the ability to respond proactively to market changes, to the ongoing improvement of laboratory best practices, and the cultivation of a work environment rooted in safety, quality, and teamwork. In sum, ADNOC Central Lab's digital transformation is not only a testament to its technical and operational leadership, but also to its enduring commitment to delivering value and driving progress in a demanding industry landscape.

References

This document is based on internal reporting and analysis of the digital transformation and operational efficiencies observed at ADNOC Central Lab, including the implementation of the Tellius Platform and associated technologies. All financial data, operational metrics, and technology descriptions are derived from laboratory performance reviews and assessments conducted by the ADNOC Central Lab team.